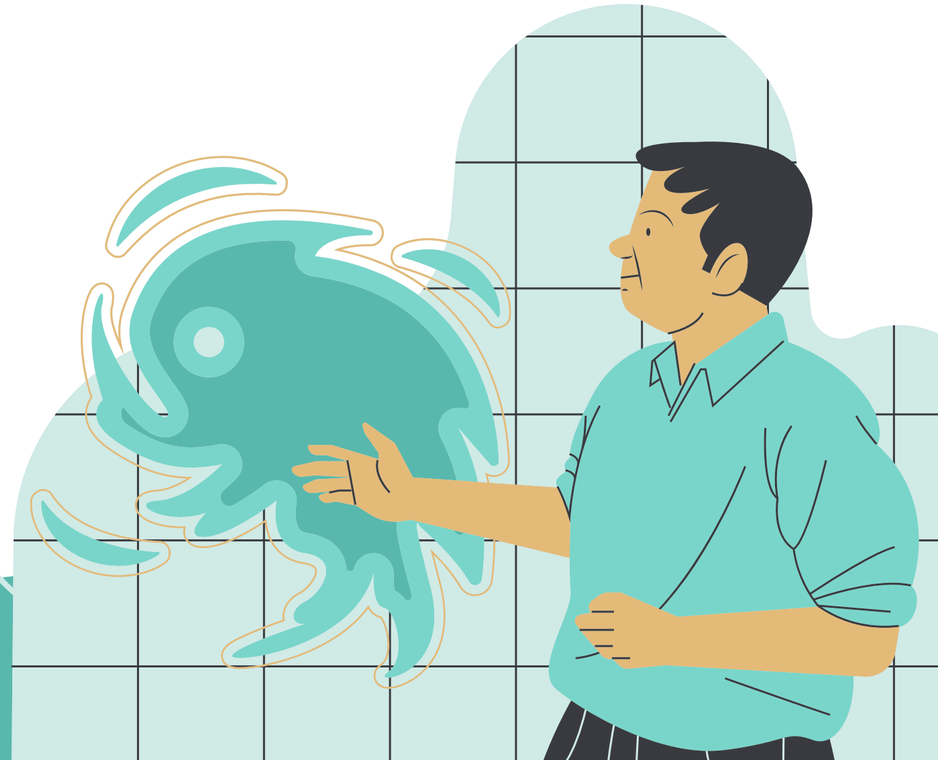
Temirlan Ilyassov
AAI-2605M · Astana IT University
AI Essentials: Theory and Application

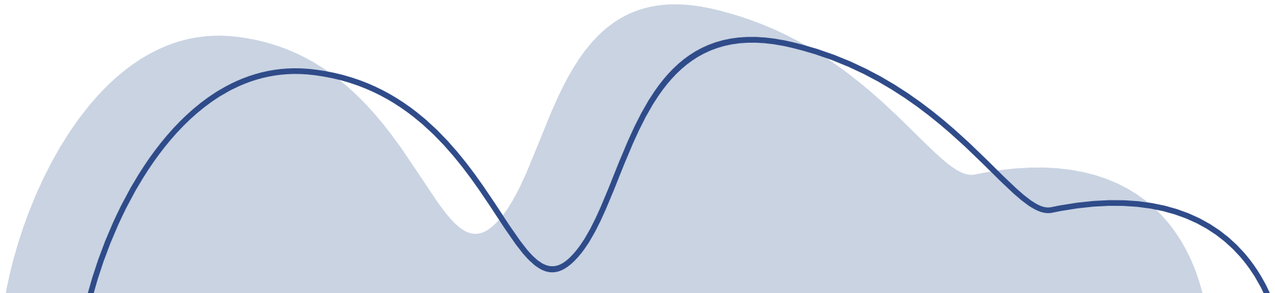
Air Quality Prediction in Almaty

Predicting PM2.5 levels using past air quality measurements



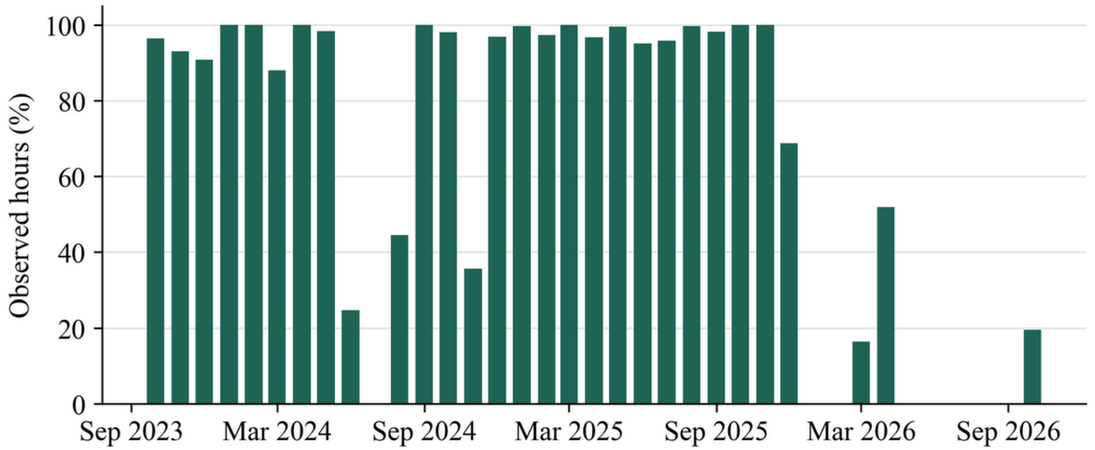
What Is The Goal?

- PM2.5 means tiny particles in the air.
- To predict the amount of these particles at a monitoring station.
- The models predict 1, 6, and 24 hours ahead.
- Question: can machine learning improve short-term PM2.5 forecasts compared with using the latest observed value?



Where Did We Get the Data?

- We used public measurements from AirData.kz.
- We studied two stations in Almaty: the US Embassy and KazHydroMet PCP #31.
- PCP #31 had 17,246 recorded hours in the saved dataset.
- About 34% of the hourly measurements were missing.

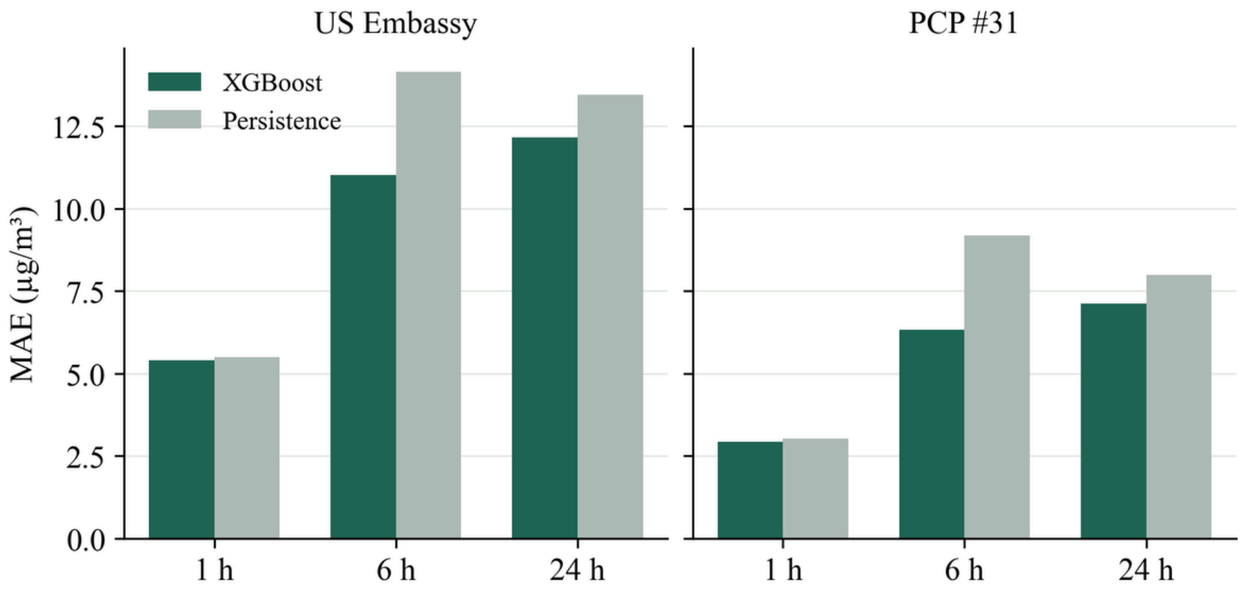


Which Models Did We Compare?

- Ridge: a simple model that learns relationships between the inputs and pollution levels.
- XGBoost: a model that combines many small decision trees.
- Latest-reading baseline: assumes the pollution level will stay the same.
- Yesterday baseline: uses yesterday's reading at the same target hour.

For a 24-hour forecast, the two baselines give the same prediction.

What Were the Main Results?

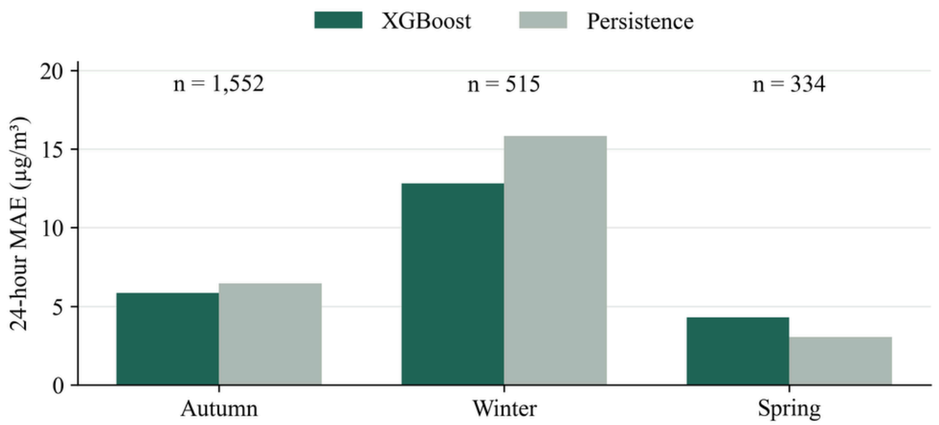


- Errors are measured in µg/m³ – micrograms of particles per cubic metre of air. Lower values are better.
- Main finding: XGBoost reduced average error by about 31% at six hours.

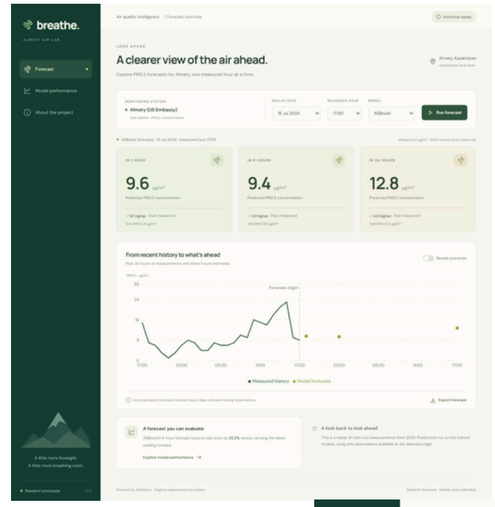
When Does the Model Make Mistakes?

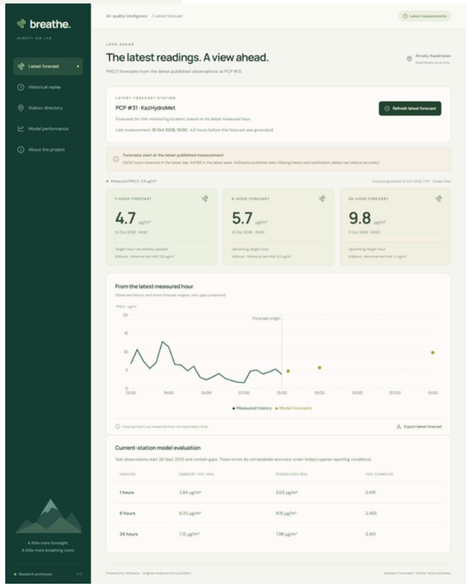
- It can predict too low during sudden pollution peaks.
- In spring, the simple latest-reading baseline performed better at PCP #31.
- With large gaps in the previous week's measurements, next-day predictions became worse.
- Good average results can hide serious mistakes in individual forecasts.

Example: One next-day prediction was 31.62 µg/m³, while the actual reading was 121.75 µg/m³.



What Does Our Application Do?





- Shows the station and its latest available measurement.
- Displays forecasts for 1, 6, and 24 hours ahead.
- Shows when the forecast starts and when the prediction applies.
- Warns users when the readings are old or incomplete.

The interface uses React and TypeScript. The prediction service uses Python and FastAPI.

Conclusion and Next Steps

- We built a working station-level forecasting system.
- Six-hour forecasts showed the strongest improvement over the simple baseline.
- We saved the datasets and settings so the experiments can be repeated.
- Next, we should test weather information, more stations, and new data.



References

- AirData.kz. (n.d.). AirDatakz OpenData: Open air quality data for Kazakhstan [Data set]. Global Shapers Almaty Hub. Retrieved October 10, 2026, from https://github.com/qazybekb/AirDatakz-OpenData
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (pp. 785–794). Association for Computing Machinery. https://doi.org/10.1145/2939672.2939785
- Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and practice (3rd ed.). OTexts. https://otexts.com/fpp3/
- scikit-learn developers. (n.d.). Ridge. scikit-learn 1.7.2 documentation. https://scikit-learn.org/1.7/modules/generated/sklearn.linear_model.Ridge.html

THANK YOU!

